Thesis Plan: CLIP-NAM for Medical Image Classification

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Project: Beyond Tabular Data: Extending Neural Additive Models to Medical Image Recognition via CLIP-Based Concept Features.

Strategy: Each phase has an internal target set ahead of the official deadline to absorb complications. No new methodology is permitted after the internal deadline of a given phase.

Deadline overview

- Fri 29 May : Replication finished (internal: Wed 27 May)
- Fri 12 Jun : Extensions and full model finished (internal: Mon 9 Jun)
- Fri 19 Jun : Draft submitted (internal: Thu 18 Jun)
- Sun 5 Jul : Final version submitted (internal: Wed 2 Jul)

1. Phase 1: Replication and Dataset Preparation

Period: 15 May to 29 May **Buffer:** 2 days

1.1 Week 1 (15 to 21 May): COMPAS replication

- **Thu 15 to Fri 16 May.** Set up the repository (PyTorch environment, version control, Colab GPU access verified). Download the ProPublica COMPAS data and run the preprocessing matching Agarwal et al. (Table 3). Skim Appendix A.5 of the NAM paper for the exact training protocol (ExU vs. ReLU choice, hidden layer sizes, 5-fold cross-validation setup).
- **Sat 17 to Mon 19 May.** Implement the base NAM architecture in PyTorch (per-feature subnetworks, summation, sigmoid). Aim for a modular design now; every extension added later should plug into this. Train a single-task NAM on COMPAS.
- **Tue 20 to Wed 21 May.** Reproduce Table 3 (single-task combined AUC of approximately 0.737). Generate the shape function plots from Figure 10 for visual verification. If AUC is off by more than about 0.01, debug rather than continue. This is the hard checkpoint: replication must work before extensions make sense.

1.2 Week 2 (22 to 29 May): Datasets and CLIP feature pipeline

- **Thu 22 to Fri 23 May.** Download HAM10000 (ISIC archive) and Chest X-ray Pneumonia (Kaggle). Verify class distributions against the proposal (about 74 percent pneumonia; about 67 percent naevi for HAM10000). Implement train/validation/test splits and the weighted cross-entropy for HAM10000.
- **Sat 24 to Sun 25 May.** Build the CLIP feature extraction pipeline. Load BiomedCLIP for chest X-rays and ViT-B/32 for HAM10000. Implement the cosine-similarity computation in equation (1) of the proposal.
- **Mon 26 May.** Construct concept sets: clinical vocabulary plus LLM-augmented candidates.

Document every prompt verbatim, as this will be needed for the sensitivity analysis later.

- **Tue 27 May (internal deadline).** Run feature extraction over both datasets, save the K-dimensional feature matrices to disk. Train a linear-probe baseline on raw CLIP embeddings to get a first benchmark number on each dataset.

- **Wed 28 to Thu 29 May (buffer).** Either catch up on slippage, or train a vanilla NAM on the medical features as a sanity check ahead of Phase 2.

2. Phase 2: Extensions and Full Model

Period: 30 May to 12 June **Buffer:** 3 days

This is where most technical risk is concentrated, so the buffer is larger.

2.1 Week 3 (30 May to 5 June): Implement the three extensions one at a time

- **Fri 30 to Sat 31 May.** Concurvity regularisation (Rügamer et al., 2023). Implement the penalty, add it to the loss, find a sensible lambda via validation. Test on COMPAS first to confirm reasonable shape functions, then move to the medical data.

- **Sun 1 to Mon 2 Jun.** Group-sparsity, the SNAM extension (Xu et al., 2022). Implement Group LASSO over sub-network parameters. Verify that uninformative concepts actually get driven to zero on a synthetic test before trusting it on real data.

- **Tue 3 to Wed 4 Jun.** Pairwise interaction terms (Kim et al., 2022). This is the most complex extension; be prepared for it to take longer than the other two. Decide upfront whether to implement from scratch or adapt the authors' code, and document the choice.

- **Thu 5 Jun.** All three extensions running together on COMPAS. Sanity check the loss curves and shape functions.

2.2 Week 4 (6 to 12 June): Run the full benchmark suite

- **Fri 6 to Sat 7 Jun.** Train the full CLIP-NAM (all three extensions) on Chest X-ray and HAM10000. Train the three baselines: LaBo, Label-Free CBM, and the raw CLIP linear probe.

- **Sun 8 Jun.** Ablation runs: remove each extension in turn. That is 3 ablations times 2 datasets, so 6 runs.

- **Mon 9 Jun (internal deadline).** All metrics compiled in a table (AUC-ROC, macro-F1, balanced accuracy for HAM10000). All top-concept shape plots generated. Decide here whether the seven-class HAM10000 task is working or whether to fall back to binary melanoma vs. rest. The proposal already names this fallback; do not wait past this date to invoke it.

- **Tue 10 to Thu 12 Jun (buffer).** Re-runs, hyperparameter touch-ups, anything that did not converge. *No new methodology after Tuesday 10 June.*

3. Phase 3: Draft Writing

Period: 13 to 19 June **Buffer:** 1 day

By the start of this phase, every number in the results section should already exist. The task here is writing only.

- **Fri 13 to Sat 14 Jun.** Methodology section. Sections 3.1 to 3.5 of the proposal already provide the scaffolding; expand them. Include the replication setup, the CLIP pipeline, and the NAM extensions with their loss equations.

- **Sun 15 Jun.** Results section: replication AUC table, benchmark comparison table, ablation table, shape function figures with discussion.
- **Mon 16 Jun.** Literature review and introduction. Most of this is in the proposal; expand and integrate.
- **Tue 17 Jun.** Discussion and conclusion. Cover identifiability, concurvity, and concept-selection limitations explicitly. These are sub-question (iv).
- **Wed 18 Jun (internal deadline).** Full draft top-to-bottom read-through. Fix structural issues, not prose-level edits.
- **Thu 19 Jun.** Final formatting pass; submit to TMS.

4. Phase 4: Revision

Period: 20 June to 5 July **Buffer:** 3 days

Supervisor feedback drives the revision. The 16-day window is structured loosely below.

- **20 to 25 Jun.** Address all supervisor feedback systematically; track it as a checklist. Rerun any experiments they ask about.
- **26 to 30 Jun.** Prose polish. Read aloud, tighten arguments, verify every claim is supported. Check mathematical notation consistency, as the assessment criteria call this out explicitly.
- **1 to 2 Jul.** References check (complete, consistent, correct, also in the rubric); figure quality; table formatting; appendix with code.
- **Wed 2 Jul (internal deadline).** Submit-ready version.
- **3 to 5 Jul (buffer).** Hold for emergencies only. Submit a day early if possible.

5. Risk register

- **Pairwise interactions (Week 3).** The most complex extension. Scope it this weekend: confirm whether Kim et al. (2022) released code, then decide implement-vs-adapt before Phase 2 begins.
- **HAM10000 seven-class task.** Fall back to binary melanoma vs. rest on 9 June if the seven-class task is not converging. The proposal already authorises this.
- **COMPAS replication off-target.** If AUC deviates by more than 0.01 on 21 May, the entire downstream plan is at risk. Do not move on without debugging.
- **Concept set quality.** Document every prompt verbatim; this is needed for the sensitivity analysis and for reproducibility credit in the rubric.